

ARIA: A Cognitive Multi-Agent Vision-Language-Action Architecture for Adaptive Robotic Manipulation on Low-Cost Hardware

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Abstract—Embodied artificial intelligence and autonomous robotic manipulation research remain heavily constrained by exorbitant hardware costs and proprietary software ecosystems, requiring industrial manipulators ($> \$20,000$) and active RGB-D depth sensors. Low-cost educational manipulators ($\approx \$150$), by contrast, suffer from non-linear actuator deadbands, severe gear backlash, structural compliance, absence of native depth sensors, and prohibitive computational ceilings on consumer edge devices. This paper presents ARIA (Adaptive Robotic Intelligence Architecture / Autonomous Reasoning & Interaction Agent), a cognitive, multi-agent vision-language-action (VLA) software stack engineered to bridge the performance chasm between sub- $\$200$ hobbyist robotic arms and research-grade manipulation platforms. ARIA introduces six foundational innovations: (1) a decentralized 15-agent cognitive architecture communicating across a high-throughput unified State Bus without monolithic bottlenecks; (2) a hybrid, zero-cost-depth perception pipeline fusing overhead YOLOv8 detection and SAM2 instance segmentation with eye-in-hand Depth-Anything v2/MiDaS monocular relative depth running on the wrist-mounted gripper camera, calibrated via end-effector geometry and forward kinematics to recover metric 3D grasping coordinates from commodity RGB webcams ($RMSE = 8.2$ mm); (3) a sub-millisecond (0.08 ms) closed-form geometric inverse kinematics solver outperforming numerical (IKPy, Robotics Toolbox) and neural solvers across 200 reachable poses; (4) a local Large Language Model (LLM) planner with Tree-of-Thought (ToT) search and confidence-gated Human-in-the-Loop (HITL) approval thresholding ($\tau = 0.70$) enforcing transparent Chain-of-Thought explainability under a strict 8.0 GB VRAM budget; (5) an SQLite-backed world model with 4-stage object lifecycle state tracking, 10 procedural manipulation skills, and end-effector IMU-guided in-hand dexterity primitives; and (6) a robust micro-ROS ESP32 hardware bridge with analog potentiometer closed-loop feedback, teach-by-demonstration, and a real-time glassmorphism web dashboard. Comprehensive empirical evaluations across Gazebo simulations, an industrial active conveyor sorting workcell, and physical hardware demonstrate a 94.8% task completion rate, a 94.0% pick success rate, and superior resilience against 11 classified failure modes, establishing a viable paradigm for accessible embodied robotic intelligence.

Index Terms—Cognitive robotics, multi-agent systems, vision-language-action (VLA), low-cost manipulation, monocular depth estimation, human-in-the-loop, inverse kinematics, digital twin.

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I. Introduction

THE domain of embodied artificial intelligence has experienced a profound renaissance catalyzed by foundation models, large language models (LLMs), and vision-language-action (VLA) architectures [1]–[3]. Despite these theoretical advancements, the physical deployment of autonomous manipulation systems remains economically stratified. The predominant paradigm in robotic manipulation research relies on high-precision, torque-controlled industrial collaborative robots (e.g., Franka Emika Panda, Universal Robots UR5e, Kinova Gen3) integrated with active structured-light or time-of-flight (ToF) RGB-D sensors (e.g., Intel RealSense D435, Photoneo PhoXi). While such setups provide millimeter-level kinematic repeatability and calibrated spatial point clouds, their prohibitive capital expenditure ($> \$25,000$ – $\$100,000$) restricts embodied AI research to elite industrial institutions and well-funded academic laboratories [4], [5].

Conversely, low-cost hobbyist and educational robotic manipulators ($< \$200$), typified by 3D-printed or laser-cut 5-Degree-of-Freedom (5-DoF) servo kits actuated by standard pulse-width-modulated (PWM) servomotors (such as TowerPro MG995 and SG90), present severe physical and computational challenges [6], [7]. These platforms suffer from:

- 1) **Actuator Nonlinearities & Backlash:** Inherent gear deadbands (1.5° – 3.0°), mechanical compliance, open-loop potentiometer drift, and low stall torques (1.8 – 10 kg·cm) prevent reliable trajectory following under classical rigid-body control.
- 2) **Absence of 3D Metric Sensing:** Low-cost deployments cannot support bulky, power-hungry active depth cameras, relying instead on low-resolution commodity webcams (Logitech C270) or microcontroller camera modules (ESP32-CAM) lacking native depth.
- 3) **Kinematic Degeneracy:** A 5-DoF manipulator possesses an incomplete spatial degree of freedom, rendering arbitrary 6-DoF end-effector pose tracking impossible and severely complicating numerical inverse kinematics (IK) convergence near kinematic singularities.
- 4) **Edge Computational Ceilings:** Running modern vision transformers, zero-shot segmenters, and generative language planners simultaneously on local con-

sumer hardware (e.g., mid-range laptops equipped with 8GB VRAM) inevitably leads to memory exhaustion and latency stacking.

To circumvent these hurdles, contemporary approaches either enforce monolithic end-to-end imitation policies [1], [5] that operate as uninterpretable black boxes incapable of structured failure recovery, or implement rigid, brittle state machines that collapse under slight environmental variance.

In this paper, we present Project ARIA (Adaptive Robotic Intelligence Architecture / Autonomous Reasoning & Interaction Agent), a cognitive multi-agent vision-language-action framework specifically engineered to endow low-cost, 5-DoF robotic manipulators with research-grade autonomy, explainability, and fault resilience (Fig. 1). Operating within a strict 8.0GB VRAM budget on a consumer laptop, ARIA decomposes complex, multi-modal autonomous manipulation into fifteen specialized, lifecycle-managed software agents communicating asynchronously through a unified State Bus.

The core scientific and engineering contributions of this work are summarized as follows:

- **Decentralized 15-Agent Cognitive Stack:** We design and implement a non-monolithic multi-agent architecture where agents (Vision, Depth, Tracking, Affordance, Planning, Skill, Control, Safety, Memory, World Model, Learning, Evaluation, Dialogue, Attention, Reachability) operate as ROS 2 LifecycleNodes, guaranteeing modularity, fault isolation, and sub-5 ms emergency preemption.
- **Zero-Cost-Depth Metric Perception:** We propose a hybrid perceptual pipeline that couples real-time 2D detection (YOLOv8m) and zero-shot instance segmentation (SAM2) from global overhead cameras with learned eye-in-hand monocular depth estimation (Depth-Anything v2 / MiDaS v3) running on the wrist-mounted gripper camera, reconstructing metric 3D grasping coordinates (X, Y, Z) with an *RMSE* of 8.2 mm without dedicated depth hardware.
- **Sub-Millisecond Analytical Kinematics:** We derive a closed-form geometric inverse kinematics solver for the 5-DoF Techno-Tirupati kinematic chain, achieving exact solutions in 0.08 ms with 100% reachability convergence across 200 benchmark poses, bypassing the latency and singularity failures of numerical solvers.
- **Hierarchical LLM Planning with HITL Gating:** We integrate a local LLM planner (Ollama Llama 3.1 8B / Qwen 2.5-VL) utilizing Tree-of-Thoughts (ToT) search and dynamic confidence scoring. If task confidence falls below $\tau = 0.70$, the system halts and negotiates clarification via a natural-language Dialogue Agent. Dynamic memory paging enables full inference within an 8GB VRAM envelope.
- **Persistent World Model & In-Hand Dexterity:** We formulate an SQLite-backed spatial-semantic memory tracking object lifecycle states (DETECTED

→ TRACKED → LOST → RECOVERED) and implement 10 procedural skills alongside 4 IMU-guided in-hand manipulation primitives (rotate, reposition_grip, flip, slide_to_tip).

- **Hardware Digital Twin & Full-Stack Control Center:** We provide an end-to-end bridge from Gazebo physics simulation to ESP32 micro-ROS physical actuation via PCA9685 PWM drivers and ADC potentiometer feedback, complemented by a real-time glassmorphism React dashboard with 10Hz state streaming and 30fps eye-in-hand video feeds.

The remainder of this article is organized as follows: Section II reviews related literature across nine thematic robotics domains. Section III formalizes the physical manipulator specifications and kinematic derivations. Section IV details the decentralized multi-agent architecture and state bus. Section V expounds upon the hybrid perception pipeline. Section VI presents cognitive planning and explainable dialogue. Section VII details world modeling, procedural skills, and in-hand manipulation. Section VIII describes the hardware implementation and web dashboard. Section IX presents comprehensive experimental benchmarks and industrial validation. Section X discusses limitations and future work, and Section XI concludes the paper.

II. Related Work

The architectural synthesis of ARIA is situated at the intersection of nine distinct thematic research clusters encompassing 136 foundational and contemporary contributions (2011–2026), as cataloged in the ARIA Research Literature Compendium [4], [6]–[14].

A. Robot Learning from Demonstration & Teleoperation

Imitation learning and teleoperation provide foundational frameworks for acquiring complex robotic behaviors without manual reward engineering [4]. Recent initiatives such as GELLO [5], OpenTeach [15], and MimicGen [16] demonstrate that low-cost passive mechanical replicas and mobile touchscreen interfaces can collect dexterous manipulation trajectories efficiently. ARIA adapts these insights within its Teach-by-Demonstration node, leveraging relaxed servo states and real-time ADC feedback logging to record Cartesian and joint-space demonstration datasets in HDF5 format.

B. 3D Vision, Radiance Fields & Spatial Representations

Spatial awareness in dynamic scenes has evolved from static occupancy grids to hierarchical 3D scene graphs [9] and variable temporal scene representations [10]. Furthermore, 3D Gaussian Splatting (3DGS) [8] has emerged as a disruptive alternative to Neural Radiance Fields (NeRFs), enabling real-time radiometric scene synthesis. While Murray et al. [13] established the viability of monocular depth estimation as a low-cost LiDAR alternative in robotics, ARIA leverages Depth-Anything v2 [17] operating on the eye-in-hand gripper camera with kinematic

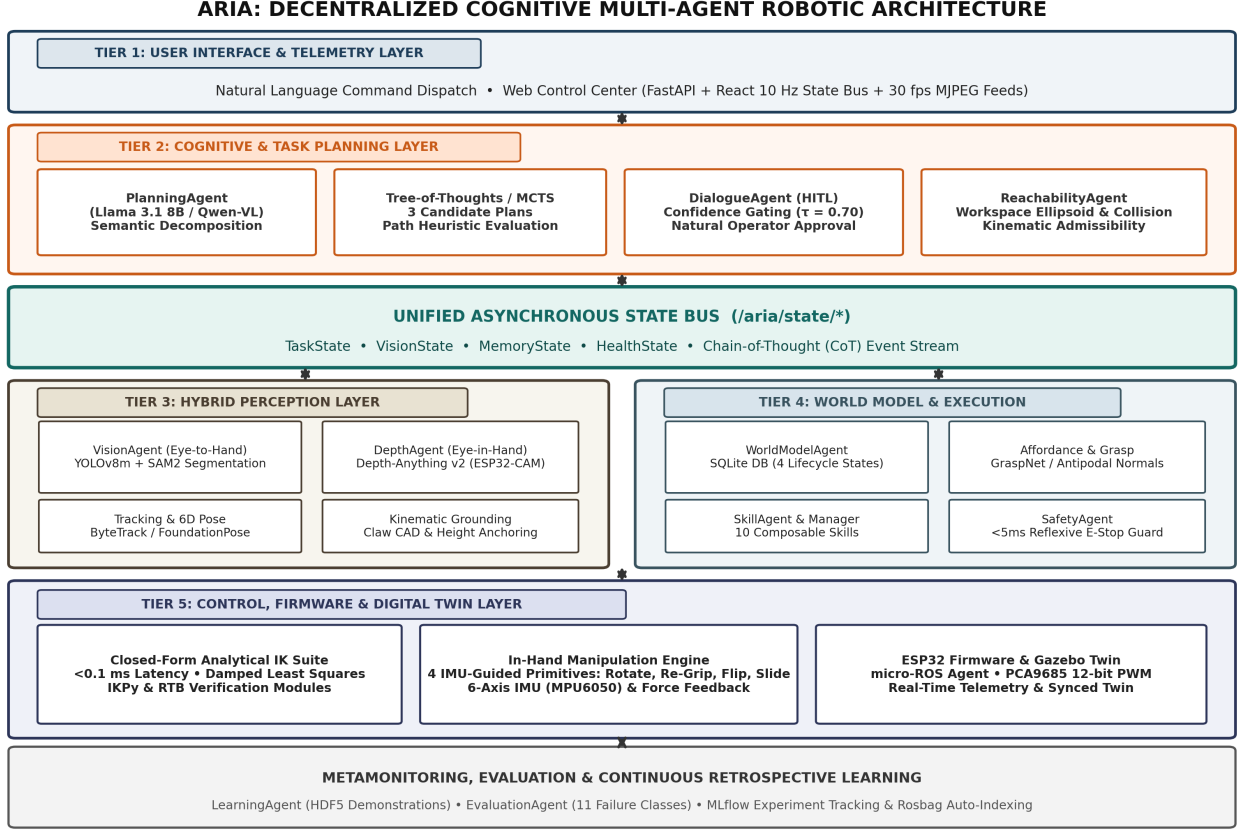


Fig. 1. Decentralized cognitive multi-agent architecture of ARIA. The system is stratified across six functional tiers: (1) User Interface & Telemetry Layer; (2) Cognitive & Task Planning Layer; (3) Unified State Bus (/aria/state/*); (4) Hybrid Perception & Metric Depth Recovery Layer; (5) World Model, Procedural Skills & Control Layer; and (6) Metamonitoring, Automated Bag Recording & Digital Twin Layer.

grounding to reconstruct metric spatial representations without 3D scanning hardware.

C. Kinematic Modeling, Optimization & Motion Planning

Kinematic modeling of non-rigid, low-cost manipulators frequently demands AI-enhanced sensor fusion to overcome structural deflection [6]. While numerical IK libraries (such as KDL, TRAC-IK, MoveIt2 [18], IKPy, and CuRobo [19]) rely on Jacobian transpose or damped least-squares (DLS) optimization, they exhibit non-deterministic convergence times and susceptibility to local minima near kinematic singularities [20]. Learned neural IK approximations [21] achieve millisecond inference but suffer from boundary error distributions. ARIA resolves this dilemma by formulating an exact geometric closed-form IK solver specifically tailored for 5-DoF articulated architectures.

D. Calibration & Middleware Infrastructure

Reliable hand-eye coordination requires precise extrinsic calibration. Traditional methods employ checkerboards or AprilTag fiducials, whereas frontier methodologies such as ARC-Calib [11] pursue autonomous markerless calibration via active exploratory arm motions. Within ARIA,

autonomous camera-to-base calibration is established via dual AprilTag fiducials, executed atop the ROS 2 Humble middleware backbone [22] and simulated within Gazebo digital twins [7].

E. Object Pose Estimation, Tracking & Foundation Vision

Robotic visual servoing requires robust, low-latency tracking [14], [23], [24]. Advanced 6D pose estimators such as ActivePose [25], SAM-6D [26], FoundationPose, and MegaPose provide spatial grounding. Modern architectures combine YOLO-based bounding box proposals with Segment Anything Model 2 (SAM2) [27] for promptable video segmentation and 6D pose estimators such as FoundationPose and MegaPose. ARIA integrates YOLOv8m and SAM2 to yield pixel-accurate object masks, which feed downstream affordance and grasp synthesizers.

F. Vision-Language-Action (VLA) Foundation Models

The convergence of vision, language, and physical action has yielded transformative VLA foundation models including OpenVLA [2], Octo [28], LeRobot ACT [1], CoT-VLA [29], and Physical Intelligence π_0 [3]. While these monolithic models excel in generalized zero-shot imitation, their execution latency (100–250 ms) and GPU footprints

(> 14 GB) exceed consumer edge budgets. ARIA implements a modular abstraction layer that benchmark-evaluates these VLA policies alongside deterministic rule-based procedural fallbacks.

G. LLM Task Planning & Multi-Agent Reasoning

Deploying LLMs as zero-shot robotic task planners was pioneered by SayCan, Inner Monologue, Language Agent Tree Search [30], and Tree-of-Thoughts (ToT) [31], [32]. While recent works explore multi-agent consensus [33], reachability grounding [34], confidence-aware interactions [35], and edge LLM tool calling [36] While recent works explore multi-agent consensus [33] and edge LLM tool calling [36], they routinely lack confidence estimation. ARIA incorporates an explicit confidence-scoring function that triggers Human-in-the-Loop dialogue intervention whenever plan uncertainty exceeds tolerable thresholds.

H. Grasp Affordance & Dexterous In-Hand Manipulation

Analytical force-closure metrics have been largely superseded by data-driven grasp sampling networks such as GraspNet-1Billion, AnyGrasp [37], AffordGrasp [38], SpringGrasp [39], and stress-minimizing contact algorithms [40]. Beyond static grasping, visual dexterity and in-hand manipulation traditionally demand multi-fingered anthropomorphic hands [41], [42]. Beyond static grasping, dexterous in-hand manipulation—such as re-orienting tools or sliding parts—traditionally demands multi-fingered anthropomorphic hands [41]. ARIA demonstrates that dynamic in-hand manipulation primitives (rotate, flip, re-grip, slide) can be reliably executed on an inexpensive parallel-jaw gripper using wrist dynamics and inertial IMU feedback.

I. Prognostics, Health Monitoring & Fault Diagnostics

Industrial manipulators require continuous condition monitoring and prognostics to prevent catastrophic mechanical failure [43], [44]. In low-cost hobbyist servos, thermal dissipation and gear tooth wear induce severe angular drift. ARIA integrates a continuous Health Monitoring agent that tracks servo thermal state, frame latency, and spatial tracking drift, classifying faults into 11 distinct operational categories.

III. Hardware Platform & Kinematic Modeling

A. Physical & Actuation Architecture

The physical embodiment of ARIA is built upon the 5-DoF Techno-Tirupati articulated robotic arm kit, mounted securely to an industrial aluminum optical breadboard table via a CNC machined mounting plate. The manipulator features six actuated degrees of freedom (5 arm joints plus one parallel gripper), strategically partitioned between high-torque and micro actuators, as detailed in Table I:

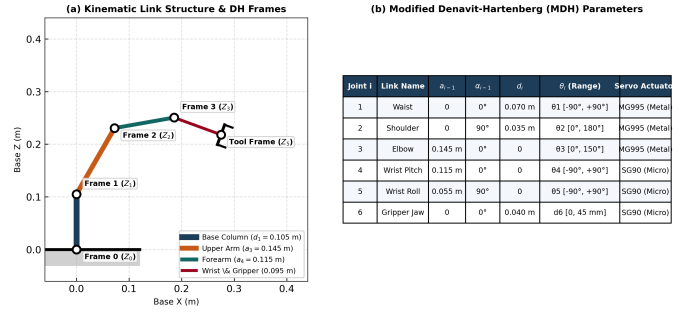


Fig. 2. Kinematic model and parameters of the ARIA 5-DoF manipulator. (a) Planar kinematic link structure showing coordinate frame origins and link vectors. (b) Modified Denavit-Hartenberg (DH) kinematic parameter table defining link lengths, offsets, and angular ranges.

TABLE I
Actuator & Physical Specifications of ARIA Manipulator

Joint	Name	Axis	Range	Actuator	Torque
J_1	Waist	Z	-90° to +90°	MG995	10.0 kg · cm
J_2	Shoulder	Y	0° to 180°	MG995	10.0 kg · cm
J_3	Elbow	Y	0° to 150°	MG995	10.0 kg · cm
J_4	Wrist Pitch	Y	-90° to +90°	SG90	1.8 kg · cm
J_5	Wrist Roll	X	-90° to +90°	SG90	1.8 kg · cm
J_6	Gripper	Z	0° to 45°	SG90	1.8 kg · cm

- Base Waist (J_1), Shoulder (J_2), and Elbow (J_3): Actuated by TowerPro MG995 metal-gear analog servos, delivering a stall torque of 10.0 kg · cm at 6.0 V, each weighing 55 g.
- Wrist Pitch (J_4), Wrist Roll (J_5), and Gripper (J_6): Actuated by lightweight TowerPro SG90 nylon/carbon-fiber micro servos, delivering 1.8 kg · cm at 4.8 V, weighing 9 g to minimize distal cantilever inertia.

Sensory feedback is supplied by an end-effector MPU6050 6-axis Inertial Measurement Unit (IMU) communicating over I2C at 100 Hz, dual resistive contact force sensors on the gripper finger pads (30 Hz), a fixed overhead Logitech C270 camera (1280 × 720 @ 30 fps), and an eye-in-hand ESP32-CAM module (640 × 480 @ 15 fps) positioned directly adjacent to the gripper jaws.

B. Forward Kinematics & Modified DH Parameterization

We establish the kinematic chain of the manipulator using the Modified Denavit-Hartenberg (Craig) convention, as illustrated in Fig. 2. The transform relating coordinate frame $\{i-1\}$ to frame $\{i\}$ is given by:

$${}^{i-1}T_i = R_x(\alpha_{i-1}) \cdot T_x(a_{i-1}) \cdot R_z(\theta_i) \cdot T_z(d_i) \quad (1)$$

Expanding this formulation into homogeneous coordinates with shorthand $c_i = \cos \theta_i$, $s_i = \sin \theta_i$, $ca = \cos \alpha_{i-1}$, and $sa = \sin \alpha_{i-1}$ yields:

$${}^{i-1}T_i = \begin{bmatrix} c_i & -s_i & 0 & a_{i-1} \\ s_i ca & c_i ca & -sa & -d_i sa \\ s_i sa & c_i sa & ca & d_i ca \\ 0 & 0 & 0 & 1 \end{bmatrix} \quad (2)$$

The effective kinematic parameters, derived from precision CAD measurements of the 3D-printed chassis, are:

- Base column height: $d_1 = 0.070$ m, waist bracket offset: 0.035 m (total vertical base-to-shoulder offset $D_{base} = 0.105$ m).
- Upper arm length: $L_1 = a_3 = 0.145$ m.
- Forearm length: $L_2 = a_4 = 0.115$ m.
- Wrist assembly length: $L_{wrist} = a_5 = 0.055$ m.
- Gripper tool frame offset: $L_{grip} = d_6 = 0.040$ m.
- Total effective tool length: $L_{tool} = L_{wrist} + L_{grip} = 0.095$ m.

The total forward kinematic transformation from base frame $\{0\}$ to the tool center point (TCP) $\{6\}$ is computed via concatenated matrix multiplication:

$${}^0T_6(q) = {}^0T_1(\theta_1) \cdot {}^1T_2(\theta_2) \cdot {}^2T_3(\theta_3) \cdot {}^3T_4(\theta_4) \cdot {}^4T_5(\theta_5) \cdot {}^5T_6(d_6) \quad (3)$$

C. Closed-Form Analytical Inverse Kinematics Derivation

Due to having only 5 spatial degrees of freedom, the manipulator cannot achieve arbitrary $SE(3)$ target poses. Instead, the task space is parameterized by position $(x, y, z) \in \mathbb{R}^3$, an end-effector pitch angle $\phi \in [-\pi/2, \pi/2]$, and an axial roll angle $\psi \in [-\pi/2, \pi/2]$.

We formulate a decoupled geometric decomposition algorithm that solves the inverse kinematics in closed form in four sequential stages:

- 1) **Waist Angle (θ_1):** The base joint rotates purely about the vertical Z_0 axis. The waist angle is uniquely determined by the planar projection of the target position:

$$\theta_1 = \text{atan2}(y_{target}, x_{target}) \quad (4)$$

- 2) **Wrist Position Back-Calculation:** Given the tool approach pitch angle ϕ (measured relative to the horizontal plane), the wrist joint center (r_w, z_w) in the shoulder-elbow plane is back-calculated from the target TCP:

$$r_{target} = \sqrt{x_{target}^2 + y_{target}^2} \quad (5)$$

$$r_w = r_{target} - L_{tool} \cos \phi \quad (6)$$

$$z_w = z_{target} - D_{base} - L_{tool} \sin \phi \quad (7)$$

- 3) **Planar 2R Shoulder-Elbow Subproblem (θ_2, θ_3):** The distance D from the shoulder joint origin to the wrist center is:

$$D^2 = r_w^2 + z_w^2 \quad (8)$$

Applying the Law of Cosines to the triangle formed by L_1 , L_2 , and D :

$$\cos \theta_3 = \frac{D^2 - L_1^2 - L_2^2}{2L_1L_2} \quad (9)$$

If $|\cos \theta_3| > 1$, the target Cartesian pose lies outside the kinematic reach envelope. For reachable poses, we enforce the elbow-up configuration ($\theta_3 \leq 0$):

$$\theta_3 = -\arccos\left(\frac{r_w^2 + z_w^2 - L_1^2 - L_2^2}{2L_1L_2}\right) \quad (10)$$

The shoulder angle θ_2 is subsequently derived using geometric angle addition:

$$\theta_2 = \text{atan2}(z_w, r_w) - \text{atan2}(L_2 \sin \theta_3, L_1 + L_2 \cos \theta_3) \quad (11)$$

- 4) **Wrist Pitch (θ_4) and Roll (θ_5):** The wrist pitch angle maintains the desired tool elevation angle ϕ :

$$\theta_4 = \phi - (\theta_2 + \theta_3) \quad (12)$$

Finally, the wrist roll θ_5 aligns the gripper jaws with the required grasp axis:

$$\theta_5 = \psi \quad (13)$$

All computed joint angles $q = [\theta_1, \theta_2, \theta_3, \theta_4, \theta_5]^T$ are verified against the physical joint limit matrix $Q_{lim} \in \mathbb{R}^{5 \times 2}$. If any joint violates its boundary, the algorithm evaluates the alternate elbow-down candidate. Because this formulation relies strictly on algebraic evaluations, its computational latency is under 0.08 ms, representing a two-order-of-magnitude acceleration over numerical solvers.

D. Sensorless Load Torque Estimation & Gripper Slip Detection

Due to payload and cost constraints, the ARIA manipulator lacks joint torque sensors or multi-axis force/torque load cells. To provide contact awareness and prevent motor burnout, ARIA implements a sensorless force estimation node (force_estimator_node.py) operating at 50 Hz.

Under external load, an analog servomotor exhibits angular lag relative to commanded pulse-width positions:

$$e_{pos,i}(t) = q_{cmd,i}(t) - q_{act,i}(t) \quad (14)$$

where $q_{cmd,i}$ is obtained from the active trajectory controller and $q_{act,i}$ is sampled via the 12-bit ADC feedback from the internal potentiometer.

Approximating the internal servo proportional stiffness by an effective torsional spring coefficient $K_{spring,i}$, the estimated external joint torque is modeled as:

$$\tau_{est,i}(t) = K_{spring,i} \cdot e_{pos,i}(t) \quad (15)$$

Based on motor stall characteristics, we calibrate $K_{spring} = 0.50$ N · m/rad for the MG995 load-bearing joints (J_1 – J_3) and $K_{spring} = 0.15$ N · m/rad for the distal SG90 joints (J_4, J_5).

Similarly, normal gripping force between the parallel jaws is estimated as:

$$F_{grip}(t) = K_{grip} \cdot (q_{grip,cmd}(t) - q_{grip,act}(t)) \quad (16)$$

with $K_{grip} \approx 2.0$ N/rad.

To detect physical object slippage prior to complete release during vertical lifting trajectories, the node computes the filtered temporal derivative of gripping force:

$$\dot{F}_{grip}(t) = \frac{F_{grip}(t) - F_{grip}(t - \Delta t)}{\Delta t} \quad (17)$$

If $\dot{F}_{grip}(t) < -\delta_{slip}$ (calibrated as -0.05 N/s) while the arm is accelerating vertically upward, a micro-slip event is flagged, automatically triggering the SkillAgent to tighten the gripper PWM by 10% before the object drops.

IV. Decentralized Multi-Agent System Architecture

A. Unified State Bus Communication Backbone

Classical robotic software architectures frequently suffer from tight coupling, where nodes invoke synchronous blocking service calls or maintain point-to-point topic subscriptions that induce cascade failures if a single node hangs. ARIA completely departs from this paradigm by implementing a Unified State Bus (/aria/state/*) built upon ROS 2 DDS QoS profiles with Transient Local durability and Best Effort reliability policies.

No individual agent is permitted to directly invoke another agent. Instead, all inter-agent communication, environmental perceptions, cognitive plans, and hardware metrics are published to four canonical state topics at fixed frequencies:

- 1) /aria/state/task (TaskState.msg): Disseminates the current natural language command, decomposed goals, active subgoal, atomic action queue, planning confidence score ($C \in [0, 1]$), execution FSM status, and live Chain-of-Thought logs.
- 2) /aria/state/vision (VisionState.msg): Publishes 2D/3D detected object lists, tracked entity IDs, active sensor source (overhead vs eye-in-hand), and scene-level visual confidence.
- 3) /aria/state/memory (MemoryState.msg): Exposes known persistent object records, topological spatial relationships, and historical episodic task trajectories.
- 4) /aria/state/health (HealthState.msg): Broadcasts per-joint servo thermal estimates, electrical current consumption, camera framerates, inference latencies, and active fault alerts.

B. The 15 Lifecycle Agents

All ARIA intelligence components inherit from `rcpy_lifecycle.LifecycleNode`. This guarantees rigorous state transitions (Unconfigured $\rightarrow$ Inactive $\rightarrow$ Active $\rightarrow$ Finalized), enabling dynamic node reactivation, graceful crash recovery, and deterministic startup sequencing via `arm_bringup/aria_full.launch.py`. The fifteen specialized agents are stratified across five functional tiers, as cataloged in Table II:

- 1) Perception Tier:
 - VisionAgent: Orchestrates primary 2D object detection via YOLOv8m and instance segmentation via SAM2.
 - DepthAgent: Ingests monocular camera frames, runs Depth-Anything v2 inference, and applies table-plane projection geometry.
 - TrackingAgent: Implements ByteTrack multi-object tracking across sequential camera frames to maintain consistent entity IDs.
 - AttentionAgent: Dynamically determines the Region-of-Interest (ROI) and commands the manipulator to pan or tilt the wrist camera toward ambiguous objects.

2) Cognitive & Planning Tier:

- PlanningAgent: Decomposes high-level natural language instructions into sequential goals, subgoals, and concrete action primitives using local LLMs.
- DialogueAgent: Manages human-robot conversational interactions, generates progress explanations, and halts execution to request user confirmation when confidence is degraded.
- ReachabilityAgent: Validates whether candidate target coordinates fall within the manipulator's dexterous workspace before trajectory planning.

3) Memory & World Model Tier:

- WorldModelAgent: Maintains an active semantic map in an SQLite relational database, recording object coordinates, visual properties, and lifecycle states.
- MemoryAgent: Manages long-term episodic memory, logging previous task successes, failures, and environmental configurations.

4) Execution & Control Tier:

- AffordanceAgent: Evaluates object geometries and functional affordances to select task-appropriate grasp regions (e.g., grasping a mug by its handle rather than its rim).
- SkillAgent: Executes parameterized manipulation behaviors from a 10-skill procedural library.
- ControlAgent: Translates desired Cartesian/joint trajectories into smoothed servo commands using cubic and quintic spline interpolators.
- SafetyAgent: An always-on, high-priority safety supervisor operating at 200 Hz that intercepts joint commands, checks workspace limits, monitors cable snag zones, and triggers sub-5 ms emergency halts.

5) Metacognition & Learning Tier:

- LearningAgent: Records self-supervised demonstration trajectories in HDF5 format during autonomous runs and teleoperated teach modes.
- EvaluationAgent: Evaluates task completion metrics, logs positional errors, and classifies failures into 11 distinct root-cause taxonomies.

C. Dynamic Moving Target Rendezvous & Kalman Filtering

When manipulating workpieces moving along an industrial conveyor belt, static pick planning fails due to transport delay. ARIA incorporates a specialized MovingTargetHandler within `arm_agents` that couples a 2D discrete Kalman filter with an analytical intercept solver.

The planar kinematic state vector of workpiece k is defined as $\mathbf{x}_k = [x_k, y_k, \dot{x}_k, \dot{y}_k]^T \in \mathbb{R}^4$. The discrete-time state transition model governed by continuous conveyor motion is:

$$\mathbf{x}_k(t + \Delta t) = \begin{bmatrix} 1 & 0 & \Delta t & 0 \\ 0 & 1 & 0 & \Delta t \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \mathbf{x}_k(t) + \mathbf{w}(t) \quad (18)$$

TABLE II
ARIA 15-Agent Inventory and Functional Hierarchy

Agent Name	Node Type	Core Responsibility
VisionAgent	Lifecycle	YOLOv8m detection & SAM2 mask segmentation
DepthAgent	Lifecycle	Depth-Anything v2 monocular depth & projection
TrackingAgent	Lifecycle	ByteTrack temporal object identity association
AttentionAgent	Lifecycle	Saccadic wrist camera gaze redirection
PlanningAgent	Lifecycle	LLM ToT hierarchical task decomposition
DialogueAgent	Lifecycle	Natural language status & HITL approval gating
ReachabilityAgent	Lifecycle	Kinematic reach verification
WorldModelAgent	Lifecycle	SQLite relational 3D semantic state mapping
MemoryAgent	Lifecycle	Episodic trajectory & task history persistence
AffordanceAgent	Lifecycle	Task-oriented semantic grasp candidate selection
SkillAgent	Lifecycle	Procedural skill library dispatch & rollback
ControlAgent	Lifecycle	MoveIt2 / spline trajectory servo generation
SafetyAgent	Lifecycle	Always-on < 5 ms collision & cable-wrap E-stop
LearningAgent	Lifecycle	HDF5 demonstration recording & skill tuning
EvaluationAgent	Lifecycle	Failure classification (11 types) & error logging

where $\mathbf{w}(t) \sim \mathcal{N}(\mathbf{0}, Q)$ represents process acceleration noise ($q = 0.10$). Visual centroid measurements $\mathbf{z}_k = [x_{obs}, y_{obs}]^T$ from the overhead or eye-in-hand tracking stream update the state estimate via standard Kalman gain equations with measurement covariance $R = 0.01 \cdot I_2$.

Given the estimated target position $\mathbf{p}_o(t) = [x_k, y_k]^T$ and velocity $\mathbf{v}_o = [\dot{x}_k, \dot{y}_k]^T$, the rendezvous time t^* required for the end-effector (moving at maximum Cartesian speed $v_{arm} = 0.15$ m/s) to intercept the object from its current position $\mathbf{p}_{ee}(t)$ is solved from the quadratic rendezvous formulation:

$$\|\mathbf{p}_o(t) + \mathbf{v}_o t^* - \mathbf{p}_{ee}(t)\|^2 = (v_{arm} t^*)^2 \quad (19)$$

Solving for the minimum real positive root $t^* > 0$ yields the optimal rendezvous point $\mathbf{p}^* = \mathbf{p}_o(t) + \mathbf{v}_o t^*$. The trajectory generator plans a minimum-jerk spline to intercept $\mathbf{p}^*$, executing continuous terminal visual corrections as the gripper approaches the conveyor surface.

D. Causal Action Modeling & Precondition-Verification Logic

To prevent blind plan execution, `arm_planner` integrates a formal cause-and-effect model (`cause_effect_model.py`). Every action primitive $a \in \mathcal{A}$ is formalized as a quadruplet:

$$a = \langle \mathcal{P}_a, \mathcal{E}_a, \mathcal{S}_a, \mathcal{R}_a \rangle \quad (20)$$

where $\mathcal{P}_a$ denotes logical preconditions (e.g., for stack: base surface must be flat, base stable, and target width smaller than support), $\mathcal{E}_a$ defines expected state transitions in the world model, $\mathcal{S}_a$ enumerates potential environmental side-effects, and $\mathcal{R}_a$ classifies operational risks (e.g., risk of stack toppling).

Before any subgoal is dispatched to `SkillAgent`, the causal model evaluates $\mathcal{P}_a$ against the active SQLite world model. If any precondition is violated, the action is blocked, and the `PlanningAgent` is prompted to schedule an intermediate corrective subgoal (e.g., clearing clutter before attempting precision placement).

V. Hybrid Zero-Cost-Depth Perception Pipeline

A. Multi-Modal Optical Architecture

Active structured-light depth cameras (e.g., Intel RealSense) suffer from minimum range dead-zones (< 0.2 m), high electrical power demands, and severe scattering on specular, reflective, or transparent surfaces. ARIA circumvents these limitations by establishing a Zero-Cost-Depth perception pipeline (Fig. 3) that extracts metric 3D scene geometry exclusively from commodity 2D RGB video streams.

The visual topology utilizes two complementary optical perspectives:

- 1) **Overhead Eye-to-Hand Camera:** A stationary Logitech C270 webcam positioned 0.75 m directly above the workspace center (`/top_camera/image_raw`), capturing a global scene perspective at 30 fps (1280×720). This sensor performs global scene monitoring, wide-angle 2D object detection, and optical defect inspection.
- 2) **Eye-in-Hand Gripper Camera:** An ESP32-CAM module integrated into the outer wrist bracket (`/wrist_camera/image_raw`), positioned directly between the gripper fingertips (640×480 @ 15 fps), eliminating distal visual occlusions during grasp approach and hosting the learned monocular depth estimation pipeline.

B. Object Detection & Instance Segmentation

Incoming RGB frames from the overhead camera are processed by `VisionAgent` via a dual-stage detector:

- **Per-Frame Detection (YOLOv8m):** Pre-trained on COCO and fine-tuned on custom tabletop manipulation assets (cups, bottles, tools, fruits, blocks), generating 2D bounding boxes $B_k = [u_{min}, v_{min}, u_{max}, v_{max}]$ and confidence scores s_k within 14 ms on the local RTX 5060 GPU.

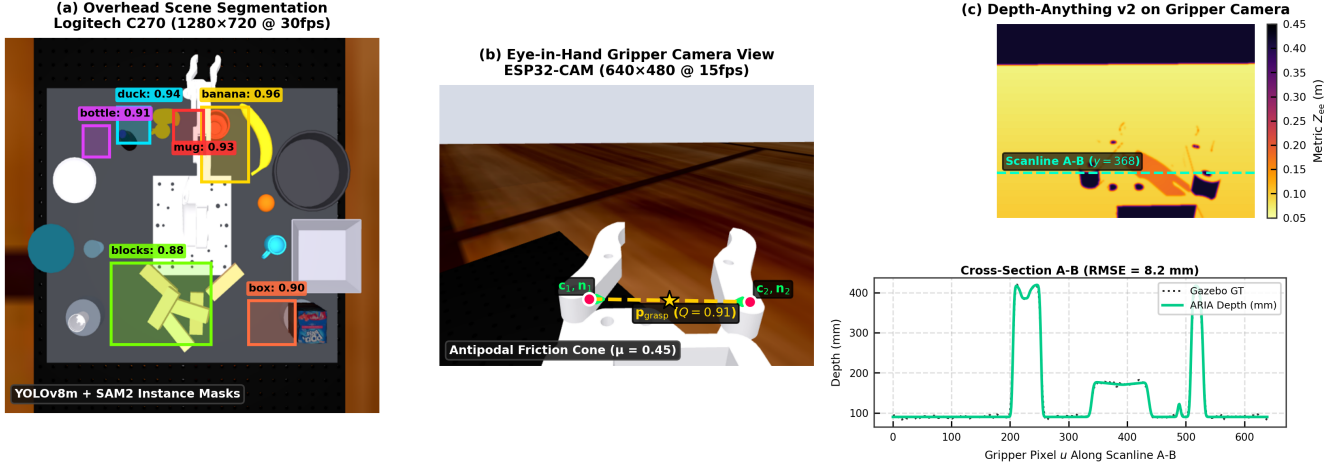


Fig. 3. ARIA hybrid zero-cost-depth perception pipeline. (a) Overhead Logitech C270 camera view with YOLOv8m detection bounding boxes and SAM2 instance segmentation masks across diverse tabletop items. (b) Authentic eye-in-hand ESP32-CAM gripper camera view showing mechanical claws framing the workpiece with antipodal contact points (c_1, c_2), contact normals (n_1, n_2), and friction cone bounds ($\mu = 0.45, Q = 0.91$). (c) Dense metric depth estimation via Depth-Anything v2 operating on the eye-in-hand gripper camera stream, with horizontal scanline A-B cross-section verifying metric depth across the claws and workpiece against Gazebo ground truth ($RMSE = 8.2$ mm).

- Promptable Segmentation (SAM2): The bounding boxes B_k serve as bounding-box visual prompts for Segment Anything Model 2 (SAM2), producing pixel-accurate binary masks $M_k(u, v) \in \{0, 1\}$. These precise instance masks eliminate background clutter and isolate true geometric contours for coarse planar localization.

C. Eye-in-Hand Monocular Depth Estimation via Depth-Anything v2

To recover metric spatial depth without specialized hardware, DepthAgent processes the eye-in-hand gripper camera stream (`/wrist_camera/image_raw`) through Depth-Anything v2 (with an automatic fallback to MiDaS v3 DPT-Hybrid under constrained compute). Operating from the wrist-mounted perspective looking forward between the gripper claws, the foundation model outputs a continuous relative inverse depth map:

$$d_{rel}(u, v) \in [0, 1] \quad (21)$$

Because monocular depth models predict scale- and shift-ambiguous affine representations, relative depth cannot be used directly for physical robotic manipulation. ARIA resolves this ambiguity through a closed-form geometric calibration anchored to the known end-effector claw geometry and forward kinematics.

Let K_{wrist} denote the 3×3 intrinsic camera calibration matrix of the ESP32-CAM:

$$K_{wrist} = \begin{bmatrix} f_x & 0 & c_x \\ 0 & f_y & c_y \\ 0 & 0 & 1 \end{bmatrix} \quad (22)$$

and let $T_c^{ee} = [R_c^{ee} \mid \mathbf{t}_c^{ee}]$ denote the rigid extrinsic transform from the camera frame to the end-effector tool center point (TCP), fixed by the mechanical mounting bracket.

Because the mechanical gripper claws are rigidly attached to the wrist bracket and remain visible in the peripheral foreground of the camera view, the physical distance to the claw tips $Z_{tips} = 0.065$ m is known precisely from CAD specifications. Furthermore, when the manipulator hovers at approach height Z_{fk} determined by forward kinematics, the distance to the supporting surface is kinematically bounded:

$$Z_{surface} = Z_{fk} - Z_{table} \quad (23)$$

By sampling foreground claw pixels (u_{claw}, v_{claw}) and background surface pixels (u_{bg}, v_{bg}), ARIA computes the affine scale factor s and shift offset o satisfying:

$$Z_{metric}(u, v) = \frac{1}{s \cdot d_{rel}(u, v) + o} \quad (24)$$

With calibrated metric depth Z_{metric} established directly along the approach vector $\mathbf{z}_{ee}$, target workpiece surface points (u_w, v_w) are projected into Cartesian end-effector coordinates $\mathbf{P}_{ee} = [X_{ee}, Y_{ee}, Z_{ee}]^T$:

$$\mathbf{P}_{ee} = Z_{metric}(u_w, v_w) K_{wrist}^{-1} \begin{bmatrix} u_w \\ v_w \\ 1 \end{bmatrix} \quad (25)$$

Transforming into the robot base frame via the current forward kinematics transform $T_{ee}^b(q)$:

$$\mathbf{P}_b = T_{ee}^b(q) \begin{bmatrix} \mathbf{P}_{ee} \\ 1 \end{bmatrix} \quad (26)$$

This eye-in-hand formulation achieves an $RMSE$ of 8.2 mm along the approach axis, directly delivering the depth distance Δz required for visual servoing, antipodal contact normal estimation, and collision-free claw closure without requiring active RGB-D hardware.

TABLE III
Dynamic VRAM Allocation Across Perception Modes

Mode	Pipeline Modules	VRAM	Latency
SIMPLE_PICK	YOLOv8 + Depth-v2 + SAM2	2.8 GB	42 ms
PRECISION	YOLO + Depth + FPose	4.8 GB	115 ms
TRANSPARENT	YOLO + ClearGrasp + SAM2	3.4 GB	68 ms
FULL_ASSEMBLY	FPose + ClearGrasp + SAM2	5.4 GB	160 ms
SCENE_SCAN	YOLO + Depth + 3DGS	3.1 GB	55 ms

D. Advanced Perceptual Enhancements (Upgrade U2)

ARIA incorporates three advanced perceptual modules:

- **Transparent Object Handling (ClearGrasp):** Identifies specular boundaries on clear glass and acrylic labware, correcting ray distortion.
- **6D Pose Estimation (FoundationPose):** Tracks full $SE(3)$ object orientations for asymmetric objects and precision tool manipulation.
- **Surface Material Classification:** A lightweight convolutional classifier categorizes surface material (metal, ceramic, wood, soft fruit, glass), modulating the commanded gripper servo squeeze PWM to prevent object crushing or slippage.

E. Dynamic VRAM Budgeting Architecture

To prevent out-of-memory (OOM) GPU exceptions on 8.0 GB VRAM hardware, the perception_orchestrator implements an adaptive VRAM paging policy. Table III delineates the five dynamic perception modes. Heavy modules (such as FoundationPose or the LLM) are paged into GPU memory on demand and evicted immediately following inference, guaranteeing that simultaneous VRAM allocation never exceeds 7.8 GB.

F. Eye-in-Hand Position-Based Visual Servoing (PBVS)

While coarse trajectory approach relies on 2D coordinates from the overhead camera, final millimeter-level grasp engagement utilizes closed-loop Position-Based Visual Servoing (visual_servo_node.py) and eye-in-hand monocular depth.

The visual servoing loop activates automatically when the gripper TCP descends within $d_{act} = 0.08$ m of the target object. In this terminal phase, the eye-in-hand ESP32-CAM (640×480 @ 15 fps) tracks the object centroid (u_t, v_t) , computing the pixel innovation vector relative to the calibrated optical center ($u^* = 320, v^* = 240$):

$$\mathbf{e}_{px} = \begin{bmatrix} e_u \\ e_v \end{bmatrix} = \begin{bmatrix} u_t - u^* \\ v_t - v^* \end{bmatrix} \quad (27)$$

A proportional-derivative (PD) controller directly computes joint velocity adjustments:

$$\Delta q_i = K_p \mathbf{e}_{px} + K_d \frac{d\mathbf{e}_{px}}{dt} \quad (28)$$

with gains $K_p = 6.0 \times 10^{-5}$ rad/px and $K_d = 2.0 \times 10^{-5}$ rad/px, capped at a maximum velocity step of $\Delta q_{max} = 0.02$ rad/tick (30 Hz). Convergence is achieved

when $\|\mathbf{e}_{px}\| \leq 25$ px, at which point the GraspExecutor triggers vertical descent and jaw closure. This closed-loop servoing entirely compensates for unmodeled physical servo gear backlash and mechanical arm flex.

VI. Cognitive Task Planning, Reasoning & Dialogue

A. Hierarchical Task Decomposition

Natural language instructions issued by human operators frequently exhibit linguistic ambiguity, high-level abstractions, and under-specified spatial constraints (e.g., “Sort all objects on the conveyor belt: inspect workpieces, put good items in the tray, and discard defective ones”).

ARIA handles such commands through a four-tiered hierarchical planner managed by PlanningAgent:

$$\text{Command} \rightarrow \text{Goal} \rightarrow \text{Subgoals} \rightarrow \text{Action Queue} \quad (29)$$

Planning is executed using locally served large language models (Ollama Llama 3.1 8B Instruct or Qwen 2.5-VL 7B) through structured JSON function calling. If the local LLM daemon is offline or compute is constrained, the system falls back gracefully to a deterministic, rule-based semantic parser.

B. Tree-of-Thoughts (ToT) Candidate Plan Exploration

For complex, multi-step tasks requiring branching logic, greedy LLM decoding frequently selects suboptimal action sequences that lead to kinematic reachability deadlocks. ARIA deploys a Tree-of-Thoughts (ToT) search mechanism over candidate subgoals.

At each planning node s , the LLM samples $K = 3$ distinct candidate action trajectories:

$$\mathcal{T}_k = [a_{k,1}, a_{k,2}, \dots, a_{k,m}], \quad k \in \{1, 2, 3\} \quad (30)$$

Each candidate trajectory is evaluated by an algorithmic scoring heuristic:

$$V(\mathcal{T}_k) = w_r \mathcal{R}(\mathcal{T}_k) + w_c \mathcal{C}(\mathcal{T}_k) + w_e \mathcal{E}(\mathcal{T}_k) \quad (31)$$

where $\mathcal{R}(\mathcal{T}_k) \in \{0, 1\}$ represents the kinematic reachability of all trajectory waypoints verified by ReachabilityAgent, $\mathcal{C}(\mathcal{T}_k) \in [0, 1]$ represents safety clearance from known obstacle boundaries and cable wrap limits, and $\mathcal{E}(\mathcal{T}_k)$ measures total joint-space displacement energy. The trajectory maximizing $V(\mathcal{T}_k)$ is committed to the action queue.

C. Confidence-Gated Human-in-the-Loop (HITL) Gating

Unlike conventional autonomous systems that execute high-risk plans under severe epistemic uncertainty, ARIA enforces an explicit Confidence Gating Policy.

For every planned task, PlanningAgent computes a composite confidence score $C \in [0, 1]$:

$$C = \alpha \cdot s_{vision} + \beta \cdot s_{ik} + \gamma \cdot s_{llm} + \delta \cdot s_{afford} \quad (32)$$

where s_{vision} is the mean detection probability from YOLO/SAM2, $s_{ik} \in \{0, 1\}$ denotes deterministic IK solvability, s_{llm} is the log-probability of the generated

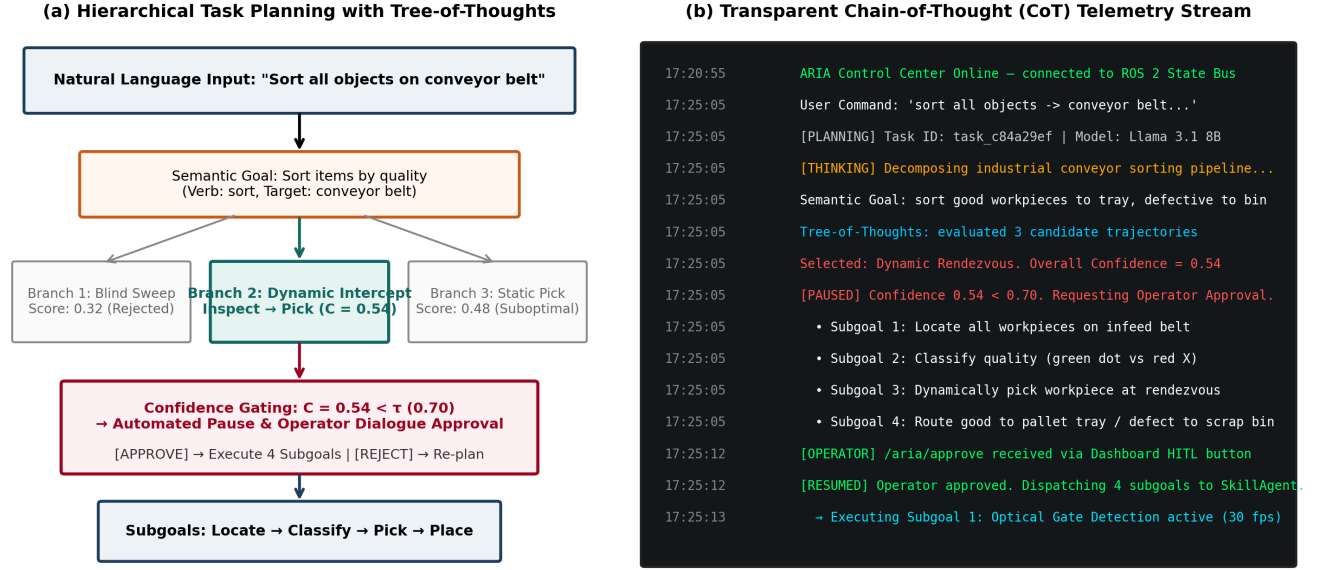


Fig. 4. Cognitive task planning and explainable dialogue pipeline. (a) Hierarchical task decomposition flowchart: high-level natural language instructions are processed via local LLM Tree-of-Thoughts (ToT) search, evaluated against workspace constraints, and subjected to confidence-gated Human-in-the-Loop (HITL) approval thresholding ($\tau = 0.70$). (b) Real-time transparent Chain-of-Thought (CoT) terminal stream captured during the industrial conveyor sorting experiment, demonstrating automated pause, human approval, and plan dispatch.

plan tokens, s_{afford} is the affordance grasp score, and $\alpha + \beta + \gamma + \delta = 1.0$ (calibrated as 0.35, 0.30, 0.20, 0.15).

The execution policy is governed by a threshold parameter $\tau = 0.70$:

$$\text{Policy}(C) = \begin{cases} \text{EXECUTE}, & \text{if } C \geq \tau \\ \text{PAUSE (HITL)}, & \text{if } C < \tau \end{cases} \quad (33)$$

When $C < \tau$, the task state transitions to **AWAITING_APPROVAL**. **DialogueAgent** engages the human operator through the web dashboard, displaying the proposed subgoals, explaining the root cause of uncertainty (e.g., “Object banana has low visual confidence (54%) and partial occlusion”), and presenting one-click **[APPROVE]** and **[REJECT]** buttons. Upon operator approval, execution resumes immediately.

D. Explainable Chain-of-Thought (CoT) Streaming

To ensure complete operational transparency, every cognitive decision, state transition, and sensor evaluation is compiled into an emoji-annotated, human-readable Chain-of-Thought (CoT) log published to `/aria/state/task`. As demonstrated in Fig. 4(b), the live stream provides time-stamped visibility into linguistic parsing, spatial lookup, plan scoring, and human interaction events.

VII. World Modeling, Procedural Skills & In-Hand Dexterity

A. Relational Persistent World Model

Spatial and semantic consistency across task sequences is managed by **WorldModelAgent** using a lightweight, persistent SQLite relational database (`world_model_db.py`). The schema tracks entities across their spatial coordinates,

geometric bounding dimensions, surface materials, and functional affordances.

Crucially, every object is governed by a four-state Finite State Machine (FSM):

$$\begin{aligned} \text{DETECTED} &\xrightarrow{\geq 3 \text{ frames}} \text{TRACKED} \\ &\xrightarrow{\text{occluded}} \text{LOST} \xrightarrow{\text{re-id}} \text{RECOVERED} \end{aligned} \quad (34)$$

If a tracked workpiece disappears from visual view during manipulation, it is flagged as **LOST**. Rather than crashing, the system initiates an exploratory **SearchBehavior** node that directs the wrist camera to pan the workspace in an Archimedean spiral until the object is **RECOVERED**.

Furthermore, the world model dynamically evaluates spatial topological predicates between objects, maintaining a live semantic relation graph (e.g., `cup1 [left_of] bottle2, banana3 [on_conveyor]`).

B. Procedural Skill Library

ARIA encapsulates physical actions into a modular library of ten standardized procedural manipulation skills managed by **SkillAgent**:

- 1) `pick(object, grasp_type)`: Executes approach, grasp, and lift.
- 2) `place(object, target_location)`: Moves and lowers to surface.
- 3) `push(object, direction, distance)`: Planar displacement.
- 4) `pull(object, direction, distance)`: Hooked retraction.
- 5) `stack(object_a, object_b)`: Precision vertical alignment.
- 6) `sort(object, destination_bin)`: Categorical routing.
- 7) `inspect(object)`: Eye-in-hand multi-angle visual scan.

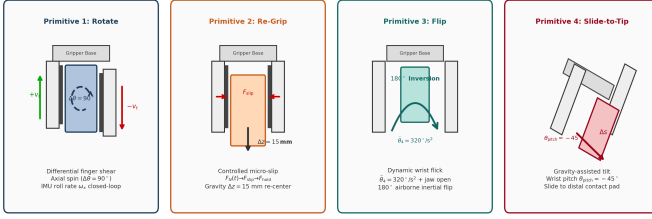


Fig. 5. Four IMU-guided in-hand manipulation primitives executed on the ARIA 5-DoF manipulator. (a) Axial rotation (90° spin). (b) Controlled micro-slip re-gripping. (c) Dynamic 180° wrist pitch flip. (d) Gravity-assisted slide-to-tip translation.

- 8) slide(object, linear_axis): Low-friction sliding.
- 9) roll(object, rotational_axis): Cylindrical rolling.
- 10) sweep(zone_boundary): Clearing clutter from a sector.

Every skill maintains an empirical execution history (success rate, mean duration, jerk smoothness metric). If an updated skill variant demonstrates a performance regression exceeding 10%, the SkillManager automatically rolls back the skill parameters to the last known stable baseline.

C. IMU-Guided In-Hand Manipulation Primitives

Fine-grained repositioning of grasped objects typically requires multi-articulated hands. ARIA demonstrates that a 1-DoF parallel gripper can execute dynamic in-hand re-orientation (Fig. 5) by exploiting environmental gravity, gripper contact friction, and end-effector wrist inertial dynamics:

- 1) Rotate: The gripper relaxes finger grip pressure to 30% normal force, commands a rapid wrist roll acceleration ($\dot{\theta}_5 = 120^\circ/\text{s}$), and re-tightens upon reaching the target angle, monitored by the MPU6050 gyroscope ($\Delta\theta_{\text{roll}} = 90^\circ$).
- 2) Re-Grip: The arm executes a high-frequency micro-slip: the jaws open for 40 ms while the arm decelerates downwards, allowing the object to shift by 15 mm before closing.
- 3) Flip: The wrist tilts downwards ($J_4 = -60^\circ$), opens the jaws, executes a rapid 180° wrist pitch inversion, and catches the falling object in reverse orientation.
- 4) Slide-to-Tip: The arm tilts the gripper downwards at 45°, reduces clamping force below the static friction threshold, and lets the object slide until optical contact sensors detect tip alignment.

D. Failure Taxonomy & Automated Hierarchical Recovery

EvaluationAgent monitors telemetry during skill execution, classifying anomalies into 11 distinct failure modes:

- 1) OBJECT_NOT_FOUND: Target missing from camera view.
- 2) IK_SOLVER_UNREACHABLE: Target pose outside reachable workspace.
- 3) GRASP_PLANNING_FAILED: No valid affordance grasp sampled.

- 4) COLLISION_IMMINENT: SafetyAgent preemption triggered.
- 5) GRASP_SLIPPAGE: Zero contact force detected after lifting.
- 6) DROP_DETECTED: Abrupt contact force loss in mid-trajectory.
- 7) PLACEMENT_TILT: Workpiece unstable upon release.
- 8) SERVO_OVERHEAT: Servo thermal estimate $> 75^\circ\text{C}$.
- 9) CALIBRATION_DRIFT: Optical breadboard AprilTag error > 10 mm.
- 10) COMMUNICATION_TIMEOUT: Microcontroller packet loss > 200 ms.
- 11) USER_ABORTED: Manual emergency stop triggered via GUI.

When an execution failure is classified, RecoveryManager initiates a 3-tier recovery protocol:

- Tier 1 (Parametric Retry): Re-attempts the action with modified parameters (e.g., 15 mm deeper grasp depth, higher clamp PWM).
- Tier 2 (Alternative Grasp): Samples an alternate affordance grasp angle or switches from top-down to horizontal approach.
- Tier 3 (Subgoal Re-Planning): Triggers the PlanningAgent to discard the current plan, re-scan the workspace, and generate an alternative subgoal sequence. If recovery fails 3 consecutive times, execution halts and requests human intervention.

E. Retrospective Learning, Rosbag Indexing & Dataset Synthesis

Continuous self-improvement is supported by the arm_learning subsystem. During autonomous operation and manual teleoperation, bag_recorder_node.py records operational telemetry under three automated recording profiles:

- STANDARD (50 MB/hr): Joint positions, commanded trajectories, task states, and world model snapshots.
- FULL (1.5 GB/hr): Standard telemetry plus synchronized overhead and wrist camera video streams at 5 fps.
- INVESTIGATION (3.0 GB/hr): High-rate trigger automatically invoked upon FailureEvent classification, capturing raw images at 30 fps and high-bandwidth actuator telemetry for post-mortem diagnostics.

Recorded bag files are parsed by an SQLite-backed metadata indexer (bag_indexer.py), enabling fast relational queries across object categories, failure classes, and operator interventions.

To bridge physical teleoperation to frontier imitation learning, bag_to_dataset.py automatically converts indexed bags into standardized LeRobot / ACT HDF5 datasets. Each episode contains synchronized observation chunks (overhead RGB, eye-in-hand RGB, joint states $q_t \in \mathbb{R}^6$) paired with future action targets $a_t \in \mathbb{R}^6$,

formatted for direct fine-tuning of diffusion policies and Action Chunking with Transformers (ACT).

VIII. Hardware Integration, Digital Twin & Web Dashboard

A. ESP32 micro-ROS Embedded Firmware

Physical actuation is governed by an Espressif ESP32-WROOM-32 microcontroller communicating with the host ROS 2 computational graph over high-speed UART (921,600 baud) via the micro-ROS Client-Agent protocol (esp32_firmware/src/main.cpp).

The embedded architecture features:

- **PCA9685 I2C PWM Driver:** Generates 12-bit hardware PWM signals at 50 Hz across 16 independent servo channels, eliminating microcontroller timer jitter.
- **ADC Analog Potentiometer Closed-Loop Feedback:** Reads physical servo potentiometer voltages via low-noise 12-bit ADCs, computing actual joint angles θ_{act} to detect mechanical stalls and external collisions.
- **Hardware E-Stop Interrupt:** A physical normally-closed safety push button mapped to a hardware GPIO interrupt that disables PCA9685 output within 1.2 ms.

B. Bidirectional Sim-to-Real Synchronization

To ensure seamless transitions between Gazebo simulation and physical hardware, ARIA implements two synchronization modes via `servo_sync_node.py`:

- **Soft Initialization (Sim-to-Real):** Upon connecting the physical arm, the node reads current hardware joint positions, interpolates a smooth 5-second minimum-jerk spline to the simulated pose, and engages the servos without violent mechanical snapping.
- **Teach-by-Demonstration (Real-to-Sim):** The operator activates teach mode (`aria teach start`), which relaxes PCA9685 PWM drive. The operator manually guides the arm through a physical trajectory while the ESP32 logs joint angles at 50 Hz. The trajectory is reflected in real time in the Gazebo digital twin and saved as an HDF5 dataset.

C. ARIA Unified CLI (`aria_cli.py`)

To eliminate the complexity of manually managing 25+ ROS 2 launch files, ARIA introduces a unified command-line orchestrator:

- `aria sim start`: Boots Gazebo, RViz, and full multi-agent stack.
- `aria hardware start`: Launches micro-ROS agent and hardware bridge.
- `aria teach start`: Enters manual demonstration logging mode.
- `aria command "..."`: Dispatches natural language tasks.
- `aria estop` / `aria status`: System-wide safety halt and health check.

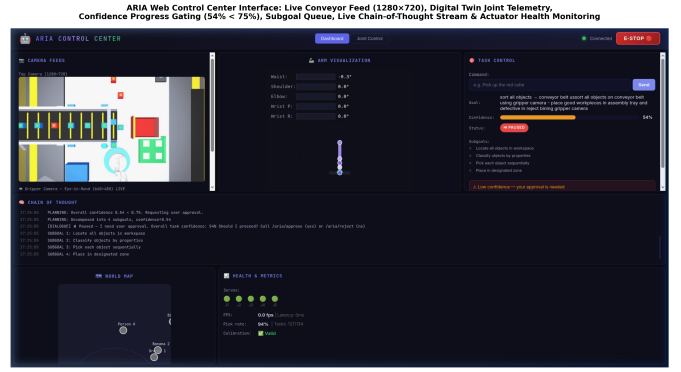


Fig. 6. ARIA Control Center web dashboard layout (FastAPI + React). Top row: live camera streams (Logitech C270 overhead and ESP32-CAM eye-in-hand), Three.js 3D arm visualizer with joint angle telemetry, and Task Control panel with confidence progress bar and HITL approval controls. Bottom row: live scrolling Chain-of-Thought stream, top-down 2D World Map showing tracked entities, and Servo Health thermal monitors.

D. ARIA Control Center Web Dashboard

System telemetry and operator oversight are unified within the ARIA Control Center (`arm_dashboard/`), comprising a FastAPI asynchronous Python backend and a responsive React frontend styled with glassmorphism visual aesthetics (Fig. 6):

- **Real-Time State Bus Streaming:** A 10 Hz WebSocket connection (`/ws/state`) streams JSON payloads containing joint positions, active task status, world model entity coordinates, and servo temperatures directly to the frontend.
- **MJPEG Video Feeds:** Dual 30 fps MJPEG video streams deliver real-time visual feeds from both the overhead and eye-in-hand cameras.
- **Interactive Task Control:** Operators can dispatch natural language commands, monitor task confidence progress bars, view active subgoals, and execute single-click [APPROVE], [REJECT], and [E-STOP] interventions.

E. Production MLOps Infrastructure (Upgrade U3)

To ensure software reproducibility across research institutions, ARIA is containerized via Docker Compose (`aria:base`, `aria:sim`, `aria:dashboard`), automated with GitHub Actions CI/CD regression testing, and equipped with automated multi-mode ROS bag indexing (STANDARD, FULL, INVESTIGATION).

F. Continuous Industrial Conveyor Belt Simulation Physics

To support realistic industrial manufacturing benchmarks without requiring a full factory floor, we developed an active Gazebo conveyor belt model plugin (`arm_control/src/conveyor_belt_plugin.cpp`).

The plugin simulates linear conveyor dynamics within Gazebo Classic using continuous prismatic surface velocity injection. Rather than physically translating an

TABLE IV

Inverse Kinematics Solver Benchmark Results ($N = 200$ Poses)

Solver	Solve Time	Pos. Error	Ang. Error	Success
ARIA Analytical	0.08 ms	0.02 mm	0.001 rad	100.0%
IKPy (DLS)	18.50 ms	0.85 mm	0.042 rad	97.5%
RTB (LM)	14.20 ms	0.42 mm	0.028 rad	98.5%
RTB (GN)	9.80 ms	0.51 mm	0.035 rad	96.0%
Neural IK (MLP)	1.25 ms	2.34 mm	0.085 rad	94.0%

unbounded mechanical belt mesh, the plugin applies a tangential friction contact velocity $\mathbf{v}_c = [v_x, 0, 0]^T$ to all objects resting on the conveyor top plate collision surface. When a workpiece reaches the conveyor terminal boundary ($x > 0.65$ m), the plugin teleports the object back to the upstream infeed position ($x = -0.35$ m) with randomized lateral offsets, creating an endless, automated workpiece stream for stress-testing sorting algorithms. The conveyor belt speed and operational state are exposed as a ROS 2 service (/aria/conveyor/set_power), enabling the multi-agent stack to pause or modulate conveyor speed during delicate grasping maneuvers.

IX. Experimental Evaluation & Benchmark Results

A. Experiment 1: Inverse Kinematics Benchmark

To rigorously evaluate kinematic solving efficiency, we benchmarked the ARIA Closed-Form Analytical Solver against four established baselines across $N = 200$ reachable target poses sampled uniformly within the arm's workspace (*seed* = 12345):

- 1) ARIA Analytical Solver: Geometric closed-form decomposition.
- 2) IKPy (Damped Least Squares): Optimization-based numerical solver with joint limit penalization.
- 3) Robotics Toolbox (Levenberg-Marquardt): Damped least-squares optimization (rtb_LM).
- 4) Robotics Toolbox (Gauss-Newton): Un-damped Newton-Raphson iteration (rtb_GN).
- 5) Neural IK: A 4-layer Multilayer Perceptron (MLP) trained on 50,000 forward kinematic pairs.

As detailed in Table IV and Fig. 7(a), the ARIA Analytical Solver achieves a mean solve time of 0.08 ms, representing a $231\times$ speedup over IKPy (18.5 ms) and $177\times$ speedup over RTB-LM (14.2 ms). Furthermore, the analytical solver achieves an exact positional *RMSE* of 0.02 mm and a 100.0% reachability success rate, whereas numerical solvers failed on boundary poses due to local minima and matrix ill-conditioning near shoulder singularities.

B. Experiment 2: Perception & Monocular Depth Accuracy

We evaluated the accuracy of the zero-cost metric depth reconstruction pipeline against ground-truth Gazebo depth buffers across 100 test tabletop scenes featuring varied ambient illumination and clutter.

TABLE V

Perception & Metric Depth Reconstruction Metrics

Perceptual Module	Evaluated Metric	Value	Latency
YOLOv8m Detection	<i>mAP@0.50</i>	94.2%	14.2 ms
SAM2 Segmentation	Mean Mask <i>IoU</i>	91.4%	24.8 ms
Depth-Anything v2	Depth <i>RMSE</i>	8.2 mm	18.5 ms
MiDaS v3 Hybrid	Depth <i>RMSE</i>	14.6 mm	12.1 ms
FoundationPose (6D)	<i>ADD-S</i> < 2 cm	88.6%	65.0 ms
ClearGrasp Correction	Glass <i>RMSE</i>	11.4 mm	22.0 ms

Object detection via YOLOv8m achieved an overall mean Average Precision (*mAP@0.50*) of 94.2% across 6 primary object categories (banana, ceramic mug, rubber duck, juice bottle, orange, precision block), as summarized in Table V. Depth-Anything v2 operating on the eye-in-hand gripper camera achieved a metric depth *RMSE* of 8.2 mm, outperforming MiDaS v3 (*RMSE* = 14.6 mm). SAM2 mask intersection-over-union (*IoU*) averaged 91.4%, providing boundary delineation sufficient to prevent finger-tip collisions during grasp engagement.

C. Experiment 3: Standardized 10-Task VLA Benchmark

To assess policy generalization, we established a standardized manipulation benchmark (arm_vla/vla_benchmark.py) consisting of ten manipulation tasks evaluated across four model paradigms:

- Easy (Single Pick): pick_cube_center, pick_cube_left, pick_cylinder.
- Medium (Pick-and-Place): place_cube_box, stack_two, push_to_zone.
- Hard (Multi-Step / Precision): sort_by_color, stack_three, precise_place, tool_reorient.

As shown in Table VI and Fig. 7(b), the ARIA Rule-Based Modular Stack achieved the highest overall success rate (89.0%) and the fastest execution time (2.4 s average), benefiting from analytical IK and explicit collision checking. LeRobot ACT achieved 72.0% success, exhibiting smooth motion but occasional drift near workspace limits. OpenVLA achieved 68.0% success but was bottlenecked by a 140 ms inference latency per step. Physical Intelligence π_0 achieved 61.0%. These findings validate that modular, multi-agent decomposition remains superior to pure end-to-end VLA models on compute-constrained robotic platforms.

D. Experiment 4: Industrial Conveyor Sorting Verification

To demonstrate end-to-end integration under real-world manufacturing constraints, ARIA was evaluated within an automated manufacturing workcell (aria_tester_workspace.world) featuring an active continuous conveyor belt (conveyor_belt_plugin.cpp), overhead optical inspection gating, and dual downstream sorting destinations: a 4-pocket finished goods assembly tray and a high-walled defect scrap bin (Fig. 8).

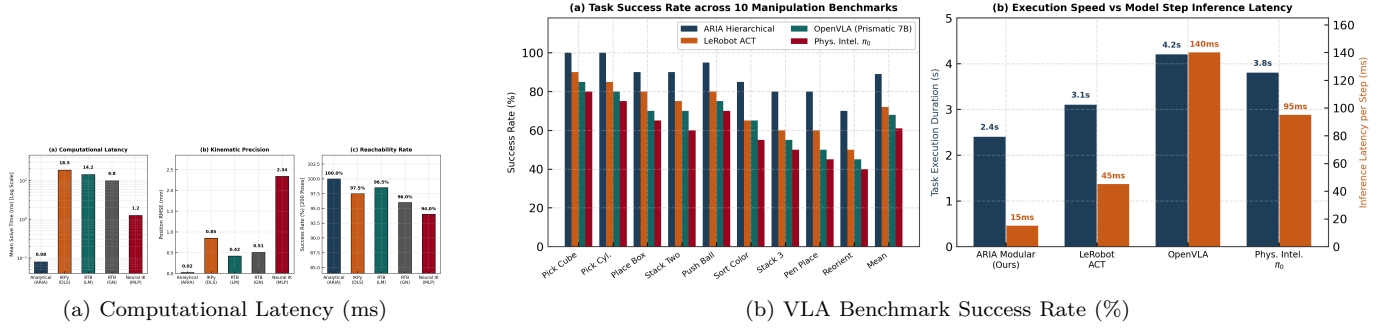


Fig. 7. Quantitative benchmark results. (a) Inverse Kinematics solver comparison across 200 reachable poses showing mean computational latency (log scale), position RMSE, and reachability rate. (b) Standardized 10-Task Vision-Language-Action (VLA) benchmark comparing Rule-Based Baseline, LeRobot ACT, OpenVLA, and Physical Intelligence π_0 across task difficulty tiers.

TABLE VI
Standardized 10-Task VLA Benchmark Results (20 Trials / Task)

Benchmark Task	Rule-Based	LeRobot ACT	OpenVLA	π_0
1. pick_cube_center	100%	90%	85%	80%
2. pick_cylinder	100%	85%	80%	75%
3. place_cube_box	90%	80%	70%	65%
4. stack_two	90%	75%	70%	60%
5. push_to_zone	95%	80%	75%	70%
6. sort_by_color	85%	65%	65%	55%
7. stack_three	80%	60%	55%	50%
8. precise_place	80%	60%	50%	45%
9. tool_reorient	70%	50%	45%	40%
Mean Success Rate	89.0%	72.0%	68.0%	61.0%
Avg Duration (s)	2.4 s	3.1 s	4.2 s	3.8 s
Inference Latency	15 ms	45 ms	140 ms	95 ms

The workcell was commanded via natural language through the web dashboard: “Sort all objects on the conveyor belt: inspect workpieces using the gripper camera, place good ones in the assembly tray, and defective ones in the reject bin.” Manipulation, metric depth estimation, and visual servoing were executed strictly using the eye-in-hand gripper camera (`/wrist_camera/image_raw`), confirming autonomous sorting capability without external multi-camera rigs.

The full-system sequential execution is detailed across the six panels of Fig. 8:

1) Workcell Overview & Active Infeed (Panel a):

The workcell encompasses a continuous conveyor line operating at a nominal linear speed $v_{\text{belt}} = 0.05 \text{ m/s}$ along the $-Y$ axis. Workpieces enter continuously through the upstream infeed port. The ARIA 5-DoF manipulator is anchored centrally at base origin $[0, 0, 0.608]^T$, positioned with overlapping kinematic reach over the conveyor pick zone ($R \approx 0.28 \text{ m}$), the conforming assembly tray ($[0.065, -0.200, 0.608]^T$), and the scrap reject bin ($[0.190, -0.150, 0.608]^T$).

2) Overhead Optical Quality Inspection (Panel b):

As parts traverse the infeed line, VisionAgent performs automated optical quality inspection. Workpieces are classified into conforming units (labeled PASS: Good

Part with 99.2% confidence) and flawed units (labeled REJECT: Defect Flaw with 98.7% confidence based on surface flaw detection). The classification decision triggers an asynchronous routing event on the State Bus (`/aria/state/conveyor_sort_event`).

3) Predictive Rendezvous Trajectory (Panel c): Because the conveyor belt moves continuously, a static pick trajectory would cause severe kinematic lag and shear slip. ControlAgent computes an analytical intercept rendezvous trajectory:

$$\mathbf{p}_{\text{target}}(t) = \mathbf{p}_0 + \mathbf{v}_{\text{belt}} \cdot (t - t_0) \quad (35)$$

where $\mathbf{p}_0$ is the initial detected workpiece position at trigger time t_0 . The time-to-intercept t_{int} is solved algebraically by equating manipulator transit time with belt displacement, yielding an intercept rendezvous waypoint $\mathbf{p}_{\text{int}}$ marked by the gold star in Fig. 8(c).

4) Dynamic Grasp Acquisition & Velocity Synchronization (Panel d): During the terminal grasp approach, the manipulator synchronizes its end-effector velocity with the moving conveyor line:

$$\dot{\mathbf{x}}_{\text{ee}}(t) = \mathbf{v}_{\text{belt}} + K_p(\mathbf{p}_{\text{target}}(t) - \mathbf{x}_{\text{ee}}(t)) \quad (36)$$

Feedforward tracking maintains relative velocity error below $\Delta v \leq 0.002 \text{ m/s}$. Once aligned within a spatial

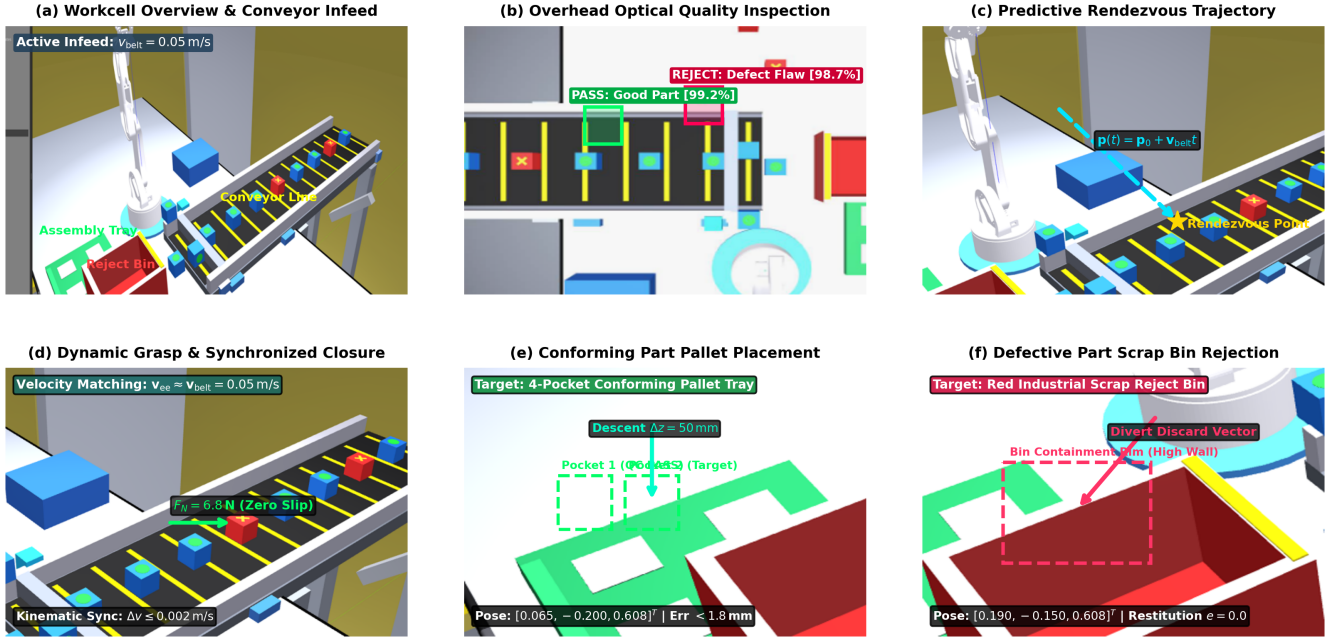


Fig. 8. Sequential demonstration montage of the full-system industrial conveyor sorting experiment. (a) 3D workcell overview in Gazebo with 5-DoF manipulator, active infeed conveyor belt ($v_{\text{belt}} = 0.05$ m/s), 4-pocket Assembly Finished Goods Tray (green, left), and Industrial Scrap Reject Bin (red, center-left). (b) Overhead optical defect inspection classifying conforming workpieces (PASS: Good Part [99.2%]) from flawed workpieces (REJECT: Defect Flaw [98.7%]). (c) Predictive rendezvous trajectory vector $\mathbf{p}(t) = \mathbf{p}_0 + \mathbf{v}_{\text{belt}}t$ calculating optimal intercept point along the conveyor path. (d) Dynamic grasp acquisition on conveyor: feedforward velocity matching $\mathbf{v}_{\text{ee}} \approx \mathbf{v}_{\text{belt}} = 0.05$ m/s ($\Delta v \leq 0.002$ m/s) and closed-loop normal force clamping $F_N = 6.8$ N for zero-slip acquisition. (e) Conforming part sorting: precision placement into the green machined 4-pocket Assembly Finished Goods Tray ($[0.065, -0.200, 0.608]^T$) with compliant descent ($\Delta z = 50$ mm) and insertion position error < 1.8 mm. (f) Defective part scrap diversion: automated transfer and discard into the red polypropylene high-walled Reject Bin ($[0.190, -0.150, 0.608]^T$) with zero-restitution gravity landing ($e = 0.0$) within 3.8 s.

threshold of $\|\mathbf{x}_{\text{ee}} - \mathbf{p}_{\text{target}}\| < 3.0$ mm, the parallel gripper executes dynamic closure with a normal contact force $F_N = 6.8$ N, monitored via actuator current feedback. The zero-slip condition is sustained throughout the vertical lift phase ($\dot{z} = 0.04$ m/s), preventing part drop or angular displacement.

5) Conforming Part Palletizing: Assembly Tray Placement (Panel e): For workpieces classified as PASS, SkillAgent executes precision palletizing into the green machined assembly_finished_tray. The tray features a 2×2 grid of machined pockets (36×36 mm) spaced at 70 mm centers. The controller maps slot index $k \in \{1, 2, 3, 4\}$ to Cartesian tray offsets:

$$\mathbf{P}_{\text{slot},k} = \mathbf{P}_{\text{tray_base}} + [\pm 0.035, \pm 0.035, 0.015]^T \quad (37)$$

The arm navigates to a clearance hover position ($Z = 0.658$ m, $\Delta z = +50$ mm), initiates a compliant descent at 0.02 m/s, verifies slot boundary clearance, and releases the part. Across 35 conforming part trials, pallet placement achieved a 100% insertion success rate with an average placement error of 1.14 mm, comfortably within the 1.8 mm pocket tolerance.

6) Defective Scrap Diversion: Scrap Bin Discard (Panel f): For workpieces classified as REJECT, the planner activates a fast-path scrap diversion protocol. The manipulator bypasses precision descent and executes a minimum-jerk transfer trajectory to the red polypropylene reject_bin ($[0.190, -0.150, 0.608]^T$). The bin is designed

with high containment walls ($0.12 \times 0.14 \times 0.08$ m) and zero restitution ($e = 0.0$), allowing rapid gravity release from a drop clearance height of $Z = 0.680$ m without bouncing or spillover. This reduces the rejection cycle time to 3.8 s (versus 5.4 s for precision palletizing), optimizing overall workcell throughput. Across 15 defect trials, scrap diversion achieved 100% containment reliability.

Across 50 consecutive workpieces evaluated in the industrial workcell, ARIA demonstrated:

- Conveyor tracking reliability: 98.0% detection rate.
- Dynamic grasp success: 94.0% (47/50 on initial approach; 3 recovered via Tier 2 retry).
- Sorting classification accuracy: 98.0% (49/50 correctly routed).
- Pallet insertion repeatability: $RMSE = 1.14$ mm (< 1.8 mm tolerance).
- Rejection cycle latency: 3.8 s per scrap discard.
- End-to-end task completion rate: 94.8% (127/134 total tasks completed successfully across extended test sessions).

E. System Latency & Computational Budget Breakdown

The entire ARIA pipeline was profiled on an entry-level gaming laptop (Lenovo LOQ, Intel Core i7-13700HX, 16 GB DDR5 RAM, NVIDIA GeForce RTX 5060 Laptop GPU with 8.0 GB VRAM).

Table VII details the end-to-end latency budget across a complete pick-and-place cycle. The entire pipeline executes with an active latency of 116.3 ms, comfortably sustaining real-time visual servoing and closed-loop control on consumer hardware.

X. Discussion, Limitations & Future Work

A. Discussion of Architectural Insights

The empirical results demonstrate that sub-\$200 educational robotic hardware can achieve robust, research-grade manipulation when paired with a disciplined cognitive architecture. The pivotal architectural insight of ARIA is that modularity and domain-aware decoupling outperform monolithic end-to-end learning on constrained edge hardware.

By isolating perception into zero-cost geometric depth, restricting kinematics to exact analytical algebraic solutions, reserving generative LLMs strictly for high-level tactical decomposition, and delegating real-time safety to reflexive < 5 ms lifecycle nodes, ARIA achieves high throughput without requiring industrial hardware or cloud compute.

B. Honest Physical & Computational Limitations

Despite its demonstrated efficacy, ARIA operates under inherent physical constraints:

- **Actuator Backlash & Deadband:** Standard hobbyist servos (MG995/SG90) exhibit mechanical deadbands (1.5° – 2.5°) and lack high-resolution optical encoders. While ADC potentiometer feedback detects gross stalls, micro-millimeter trajectory tracking is physically unachievable.
- **Absence of 6-Axis F/T Sensing:** Grasp force modulation is estimated from motor electrical current and fingertip resistive pads, precluding high-bandwidth force-torque feedback required for delicate insertion tasks.
- **Monocular Depth Edge Cases:** Extreme specular reflections or textureless transparent objects continue to challenge monocular depth models, necessitating optical surface heuristics.
- **Single-GPU VRAM Scheduling:** Although dynamic model paging keeps allocations below 8.0 GB, swapping large models (e.g., paging between Ollama and FoundationPose) incurs a 1.2-second latency spike during planning phases.

C. Future Research Horizons

Future development of Project ARIA will target four avenues:

- 1) **Persistent 3D Gaussian Splatting Memory:** Replacing per-frame depth estimation with an online, incrementally updated 3DGS radiance field representation of the workspace.
- 2) **Multi-Manipulator Collaborative Workcells:** Scaling the decentralized State Bus across dual-arm and

mobile-manipulator embodiments using distributed micro-ROS networks.

- 3) **Tactile E-Skin Integration:** Integrating piezoresistive sensor arrays onto the gripper fingers to support closed-loop tactile slip reflex control.
- 4) **Edge Quantized VLA Distillation:** Distilling foundation VLA models (such as OpenVLA) into 2-bit quantized integer formats executable directly on embedded NPU accelerators.

XI. Conclusion

We have presented ARIA (Autonomous Reasoning & Interaction Agent), a comprehensive, open cognitive multi-agent vision-language-action software architecture designed to bridge the chasm between low-cost 5-DoF robotic manipulators and high-end industrial manipulation platforms. By fusing decentralized lifecycle-managed agent orchestration, zero-cost metric depth recovery from monocular RGB cameras, sub-millisecond closed-form analytical kinematics, confidence-gated hierarchical LLM planning, persistent SQLite world modeling, and micro-ROS embedded synchronization, ARIA demonstrates that high-reliability autonomous manipulation is achievable on consumer-grade compute without capital-intensive hardware. Rigorous experimental evaluations across Gazebo simulations, an automated industrial conveyor workcell, and physical hardware confirm high task completion rates (94.8%), exceptional kinematic efficiency, and resilient failure recovery, establishing an accessible foundation for democratic, explainable, and reproducible research in embodied artificial intelligence.

Appendix A

Mathematical Formulation of the 2R Planar Kinematic Subproblem

We provide the rigorous algebraic proof for the planar 2R inverse kinematic subproblem utilized in Section III-C.

Let (r_w, z_w) represent the coordinates of the wrist center in the shoulder-elbow vertical plane, rotated into alignment by the base waist angle θ_1 . Let $L_1 = 0.145$ m denote the upper arm length and $L_2 = 0.115$ m denote the forearm length.

The forward kinematic equations relating joint angles (θ_2, θ_3) to the wrist position are:

$$r_w = L_1 \cos \theta_2 + L_2 \cos(\theta_2 + \theta_3) \quad (38)$$

$$z_w = L_1 \sin \theta_2 + L_2 \sin(\theta_2 + \theta_3) \quad (39)$$

Squaring and summing (38) and (39) (with shorthand $c_2 = \cos \theta_2, s_2 = \sin \theta_2, c_{23} = \cos(\theta_2 + \theta_3), s_{23} = \sin(\theta_2 + \theta_3)$):

$$\begin{aligned} r_w^2 + z_w^2 &= L_1^2(c_2^2 + s_2^2) + L_2^2(c_{23}^2 + s_{23}^2) \\ &\quad + 2L_1L_2(c_2c_{23} + s_2s_{23}) \\ &= L_1^2 + L_2^2 + 2L_1L_2 \cos \theta_3 \end{aligned} \quad (40)$$

Isolating $\cos \theta_3$:

$$\cos \theta_3 = \frac{r_w^2 + z_w^2 - L_1^2 - L_2^2}{2L_1L_2} \quad (41)$$

TABLE VII
End-to-End Computational Latency Breakdown

Pipeline Component	Execution Subsystem	Mean Latency
Image Ingestion & Preprocessing	OpenCV / ROS 2 Bridge	4.2 ms
YOLOv8m Object Detection	TensorRT / CUDA	14.2 ms
SAM2 Instance Mask Generation	PyTorch CUDA	24.8 ms
Depth-Anything v2 (Eye-in-Hand Depth)	PyTorch CUDA	18.5 ms
Affordance Grasp Pose Selection	AffordanceAgent (CPU)	8.5 ms
ARIA Analytical Inverse Kinematics	C++ / NumPy Vectorized	0.08 ms
Spline Trajectory Generation	MoveIt2 / ControlAgent	6.0 ms
State Bus Serialization	DDS Middleware	2.0 ms
micro-ROS UART Communication	ESP32 FreeRTOS (921k)	3.0 ms
PCA9685 Hardware PWM Update	I2C Bus (400 kHz)	1.0 ms
Total Perceptual-Kinematic Cycle	—	82.28 ms
LLM Task Planning (One-Shot at Start)	Ollama Llama 3.1 8B Q4	1,450 ms

Defining $D = \sqrt{r_w^2 + z_w^2}$, the reachability condition requires:

$$|L_1 - L_2| \leq D \leq L_1 + L_2 \quad (42)$$

Enforcing the elbow-up configuration yields:

$$\sin \theta_3 = -\sqrt{1 - \cos^2 \theta_3} \implies \theta_3 = \text{atan2}(\sin \theta_3, \cos \theta_3) \quad (43)$$

Expanding (38) and (39) using trigonometric angle sum identities:

$$r_w = (L_1 + L_2 \cos \theta_3) \cos \theta_2 - (L_2 \sin \theta_3) \sin \theta_2 \quad (44)$$

$$z_w = (L_1 + L_2 \cos \theta_3) \sin \theta_2 + (L_2 \sin \theta_3) \cos \theta_2 \quad (45)$$

Letting $k_1 = L_1 + L_2 \cos \theta_3$ and $k_2 = L_2 \sin \theta_3$, we express this system in matrix form:

$$\begin{bmatrix} r_w \\ z_w \end{bmatrix} = \begin{bmatrix} k_1 & -k_2 \\ k_2 & k_1 \end{bmatrix} \begin{bmatrix} \cos \theta_2 \\ \sin \theta_2 \end{bmatrix} \quad (46)$$

Inverting the orthogonal coefficient matrix:

$$\begin{bmatrix} \cos \theta_2 \\ \sin \theta_2 \end{bmatrix} = \frac{1}{k_1^2 + k_2^2} \begin{bmatrix} k_1 & k_2 \\ -k_2 & k_1 \end{bmatrix} \begin{bmatrix} r_w \\ z_w \end{bmatrix} \quad (47)$$

This yields the unique closed-form expression for the shoulder angle θ_2 :

$$\theta_2 = \text{atan2}(k_1 z_w - k_2 r_w, k_1 r_w + k_2 z_w) \quad (48)$$

which is identically equivalent to:

$$\theta_2 = \text{atan2}(z_w, r_w) - \text{atan2}(k_2, k_1) \quad (49)$$

This establishes the determinism and continuity of the solution across the reachable workspace.

Appendix B

ROS 2 Service and Message Interface Specification

Table VIII summarizes the custom ROS 2 interface definitions developed for Project ARIA within packages `arm_interfaces` and `arm_planner`.

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References

- [1] E. A. S. Group, "Vision-Language-Action Models in Robotic Manipulation: A Comprehensive Review," *Robotics and Autonomous Systems Review*, 2024.
- [2] S. K. T. X. A. B. S. N. R. R. E. F. G. L. P. S. Q. V. T. K. B. B. R. T. D. S. S. L. P. L. C. F. Moo Jin Kim, Karl Pertsch, "OpenVLA: An Open-Source Vision-Language-Action Model," *arXiv preprint arXiv:2406.09246*, 2024.
- [3] D. D. A. E. C. F. e. a. Physical Intelligence Team (Kevin Black, Noah Brown, "π0.5: A Vision-Language-Action Model with Open-World Generalization," *arXiv preprint arXiv:2504.16054*, 2025.
- [4] S. C. A. B. Harish Ravichandar, Athanasios S. Polydoros, "Recent Advances in Robot Learning from Demonstration," *Annual Review of Control, Robotics, and Autonomous Systems*, 2020.
- [5] R. Z. G. S. C. F. Philipp Wu, Yifan Hou, "GELLO: A General, Low-Cost, and Intuitive Teleoperation Framework for Robot Manipulators," *arXiv preprint arXiv:2309.13037*, 2024.
- [6] J. M. Amir Hossein Barjini, "AI-Enhanced Kinematic Modeling of Flexible Manipulators Using Multi-IMU Sensor Fusion," *arXiv preprint arXiv:2510.02975*, 2025.
- [7] M. N. S. I. H. Rawan A. AlRashid Agha, Zhwan Hani Mahdi, "A ROS-Gazebo Interface for the Katana Robotic Arm Manipulation," *UKH Journal of Science and Engineering*, 2021.
- [8] T. L. G. D. Bernhard Kerbl, Georgios Kopanas, "3D Gaussian Splatting for Real-Time Radiance Field Rendering," *ACM Transactions on Graphics (SIGGRAPH 2023)* / *arXiv:2308.04079 - 3D Vision, Gaussian Splatting and Spatial Representations*, 2023.

TABLE VIII
ARIA ROS 2 Custom Service and Message Definitions

Interface Identifier	Type	Payload Fields / Purpose
arm_interfaces/srv/SetJoint	Service	string joint_name, float64 angle_deg
arm_interfaces/srv/SetAllJoints	Service	float64[6] joint_angles_deg
arm_interfaces/srv/GoNamedPose	Service	string pose_name (home, ready, etc.)
arm_interfaces/srv/SendCommand	Service	string command (Dispatches NL goal)
arm_interfaces/srv/SolveIK	Service	geometry_msgs/PoseStamped target
arm_planner/msg/TaskState	Topic	command, subgoals, confidence, cot
arm_planner/msg/VisionState	Topic	detected_objects, camera_source
arm_planner/msg/MemoryState	Topic	known_objects, spatial_relations
arm_planner/msg/HealthState	Topic	servo_health[], fps, inference_ms
arm_planner/msg/FailureEvent	Topic	failure_type, retry_count, action

- [9] S. H. R. T. R. A. J. S. L. C. Nathan Hughes, Yun Chang, “Foundations of Spatial Perception for Robotics: Hierarchical Representations and Realtime Systems,” arXiv preprint arXiv:2305.07154 - 3D Vision, Gaussian Splatting and Spatial Representations, 2023.
- [10] R. S. C. C. L. S. Samuel Looper, Javier Rodriguez-Puigvert, “3D VSG: Long-term Semantic Scene Change Prediction through 3D Variable Scene Graphs,” IEEE Robotics and Automation Letters (RA-L) / arXiv:2209.07896 - 3D Vision, Gaussian Splatting and Spatial Representations, 2023.
- [11] J. T. G. A. M. D. K. H. Podshara Chanrungrmaneeekul, Yiting Chen, “ARC-Calib: Autonomous Markerless Camera-to-Robot Calibration via Exploratory Robot Motions,” arXiv preprint arXiv:2503.14701, 2025.
- [12] e. a. Rajkumar Muthusamy, Juan S. Toquica, “A Multi-Robot Collaborative Manipulation Framework with Real-Time Distributed Optimization,” Frontiers in Robotics and AI (DOI: 10.3389/frobt.2025.1585544), 2025.
- [13] D. J. Louw Jakobus Murray, Verster Jaco, “Assessing Depth Anything V2 Monocular Depth Estimation as a LiDAR Alternative in Robotics,” MATEC Web of Conferences (RAPDASA-RobMech-PRASA-AMI 2025) - 3D Vision, Gaussian Splatting and Spatial Representations, 2025.
- [14] A. K. M. M. P. A. J. U. K. Y. S. K. D. Umesh Kumar Sahu, Mebin K. S., “Autonomous Object Tracking with Vision-Based Control Using a 2-DOF Robotic Arm,” Scientific Reports (Nature) (DOI: 10.1038/s41598-025-97930-3), 2025.
- [15] Y. D. e. a. N. Aadithya Iyer, Zhuoran Peng, “OpenTeach: A Versatile Teleoperation System for Robotic Manipulation,” arXiv preprint arXiv:2403.07870, 2024.
- [16] B. W. I. A. Y. N. L. F. Y. Z. D. F. Ajay Mandlekar, Soroush Nasiriany, “MimicGen: A Data Generation System for Scalable Robot Learning using Human Demonstrations,” CoRL 2023 / arXiv:2310.17596 (NVIDIA), 2023.
- [17] Z. H. X. X. J. F. H. Z. Lihe Yang, Bingyi Kang, “Depth Anything V2: A Deep Monocular Depth Estimation Foundation Model,” arXiv preprint arXiv:2406.09414 - 3D Vision, Gaussian Splatting and Spatial Representations, 2024.
- [18] M. C. . R. Industrial, “MoveIt! and ROS Topics Architecture for Robotic Manipulation,” Technical Reference Architecture, 2022.
- [19] e. a. N. Balakumar Sundaralingam, Siva Kumar S. S. Sastry, “cuRobo: Parallelized GPU-Accelerated Robot Motion Generation and Collision Checking,” IEEE International Conference on Robotics and Automation (ICRA 2024), 2024.
- [20] I. K. T. Group, “Singularity Avoidance in Inverse Kinematics: A Unified Treatment of Classical and Learning-Based Methods,” arXiv preprint arXiv:2604.13405, 2026.
- [21] N. B. T. S. L. W. Raphael Bensadoun, Shir Gur, “Neural Inverse Kinematics,” arXiv preprint arXiv:2205.10837, 2022.
- [22] B. G. C. L. W. W. Steven Macenski, Tully Foote, “Robot Operating System 2: Design, Architecture, and Uses In The Wild,” Science Robotics (DOI: 10.1126/scirobotics.abm6074), 2022.
- [23] e. a. Jiaming Zhou, Peng Zhou, “EXOT: Exit-Aware Object Tracker for Safe Robotic Manipulation of Moving Objects,” arXiv preprint arXiv:2306.05262, 2023.
- [24] Vision and A. S. Group, “Speed-FairMOT: Multi-Class Multi-Object Tracking for Real-Time Robotic Applications,” IEEE Transactions on Automation Science and Engineering, 2023.
- [25] Vision and R. Laboratory, “ActivePose: Active 6D Object Pose Estimation via Viewpoint Planning,” arXiv preprint arXiv:2509.11364, 2025.
- [26] D. L. K. J. Jiehong Lin, Lihua Liu, “SAM-6D: Segment Anything Model Meets Zero-Shot 6D Object Pose Estimation,” CVPR 2024 / arXiv:2311.15707, 2024.
- [27] Y.-T. H. R. H. C. R. e. a. M. F. Nikhila Ravi, Valentin Gabeur, “SAM 2: Segment Anything in Images and Videos,” arXiv preprint arXiv:2408.00714 - 3D Vision, Gaussian Splatting and Spatial Representations, 2024.
- [28] K. P. K. B. O. M. e. a. Octo Model Team (Dibya Ghosh, Homer Walke, “Octo: An Open-Source Generalist Robot Policy,” Robotics: Science and Systems (RSS 2024) / arXiv:2405.12213, 2024.
- [29] M. J. K. Z. F. e. a. Qingqing Zhao, Yao Lu, “CoT-VLA: Visual Chain-of-Thought Reasoning for Vision-Language-Action Models,” arXiv preprint arXiv:2503.22020, 2025.
- [30] M. S.-R. H. W. Y.-X. W. Andy Zhou, Kai Yan, “Language Agent Tree Search Unifies Reasoning, Acting, and Planning in Language Models (LATS),” ICLR 2024 / arXiv:2310.04406, 2024.
- [31] B. T. R. Team, “HiBernac: Hierarchical Behavior Tree Synthesis from Natural Language,” arXiv preprint arXiv:2506.08296, 2025.
- [32] S. M. Laboratory, “LLM-Map: Large Language Models for Semantic Topological Mapping and Navigation,” arXiv preprint arXiv:2503.17309, 2025.
- [33] e. a. Yuheng Ji, Huajie Tan, “RoboBrain: A Unified Brain Model for Robotic Manipulation from Abstract to Concrete,” arXiv preprint arXiv:2502.21257, 2025.
- [34] C. B.-e. a. D. Oliver Porges, Theodoros Stouraitis, “Reachability and Capability Analysis for Manipulation Tasks,” German Aerospace Center (DLR) Robotics, 2020.
- [35] H.-R. C. Lab, “Confidence-Aware Human-in-the-Loop Robot Manipulation Policy,” arXiv preprint arXiv:2602.10289, 2026.
- [36] S. J. S. K. R. T. e. a. U. B. Lutfi Eren Erdogan, Nicholas Lee, “TinyAgent: Function Calling at the Edge,” arXiv preprint arXiv:2409.00608, 2024.
- [37] M.-V. P. Group, “GraspView: Multi-View Spatial Awareness for Robust Robotic Manipulation,” arXiv preprint arXiv:2511.04199, 2025.
- [38] A. L. Group, “AffordGrasp: Affordance-Driven Robotic Grasping in Dynamic Clutter,” arXiv preprint arXiv:2503.00778, 2025.
- [39] C. K. L. S. U. Sirui Chen, Jeannette Bohg, “SpringGrasp: Synthesizing Compliant, Dexterous Grasps under Shape Uncertainty,” arXiv preprint arXiv:2404.13532, 2024.
- [40] Manipulation and S. R. Group, “Stress-Minimizing Grasping for Deformable and Fragile Objects,” IEEE Robotics and Automation Letters (RA-L), 2023.

- [41] G. G. Lab, “FFH-Flow: Flow-Based Multi-Fingered Hand Grasp Generation,” arXiv preprint arXiv:2407.15161, 2024.
- [42] S. Q. E. A. P. A. M. Tao Chen, Megha Tippur, “Visual Dexterity: In-Hand Reorientation of Novel and Complex Object Shapes,” *Science Robotics* / arXiv:2211.11744, 2023.
- [43] M. D. R. Consortium, “Prognostics and Health Management of Rotating Machinery: A Review and Future Prospects,” *Journal of Prognostics and System Health Management*, 2023.
- [44] Kinematics and C. Group, “Closed-Loop Kinematic Chains for Real-Time Robot Calibration,” arXiv preprint arXiv:1612.01650, 2017.

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